

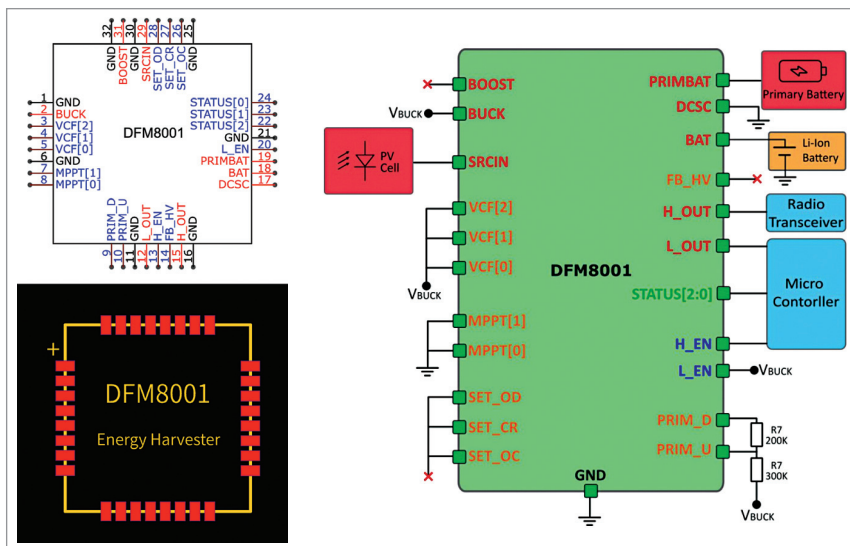


Fig. 3: Chip pinouts and application circuit (Source: DFM8001 datasheet)

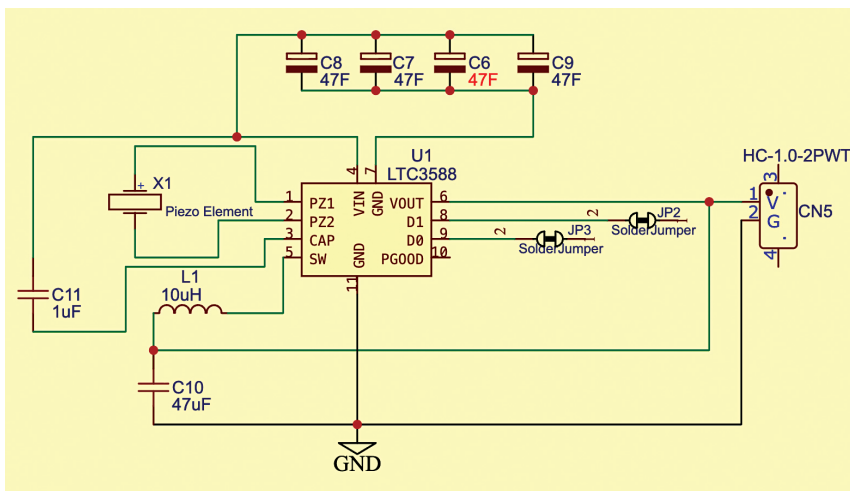


Fig. 4: LTC3588 energy-harvesting batteryless power system

power harvested from a PV cell and intelligently distributes it to power the microcontroller and radio transceiver while simultaneously charging a battery backup. This enables continuous, autonomous, long-life operation of low-power IoT systems without frequent battery replacements. Fig. 3 shows the chip pinouts and application circuit of the DFM8001.

The energy-harvesting system supports integration with a wide range of ambient energy sources. The connector enables the integration of thermoelectric generators, PV cells, piezoelectric devices, or RF energy-harvesting modules.

The SRCIN pin on the IC module, attached to connector C4, allows an ambient energy input source for harvesting. The IC then provides DSC pins, supported by supercapacitors to store the harvested energy.

According to the datasheet, the IC offers two battery options: a primary battery and a smaller rechargeable battery for additional energy storage. At the BAT pin, either a capacitor or a small lithium-ion rechargeable battery may be connected to store harvested energy. The primary battery serves as a backup source and may be rechargeable or non-rechargeable, providing power when harvested and stored

energy is insufficient. This enhances system reliability.

In this design, a capacitor is connected to the BAT pin for storage, although users may replace it with a small lithium-ion battery, if required. The system also provides load connections to HEN, LEN, or GND, depending on the load's operating-voltage requirements.

Construction and testing of DFM8001-based circuit. As shown on the left side of Fig. 2, the circuit is centred on the DFM8001 power-management IC, which efficiently handles energy-harvesting input and dual-battery sources. The SRCIN connector supports piezo, solar, or RF harvesters using MPPT for optimal extraction, while the primary battery connectors interface with non-rechargeable and rechargeable batteries. Capacitors C1 through C5 ensure stable charging and minimal ripple, and the load connector provides regulated power to IoT modules. JP1 allows enabling or isolating the load during setup.

Testing involves powering the board through each available source to verify automatic switching and stable voltage output. The load output and status indicators confirm correct operation, ensuring uninterrupted load performance under varying harvesting and battery conditions.

LTC 3588 energy harvesting system

Fig. 4 shows the circuit diagram of the LTC3588 energy-harvesting system. It is built around the LTC3588 IC, supercapacitors, and

TABLE 2 CONFIGURATIONS OF D0 AND D1 OF LTC3588				
D1	D0	V _{OUT}	V _{OUT}	Quiescent Current (I _{VOUT})
0	0	1.8V		44nA
0	1	2.5V		62nA
1	0	3.3V		81nA
1	1	3.6V		89nA

(Source: LTC3588 datasheet)